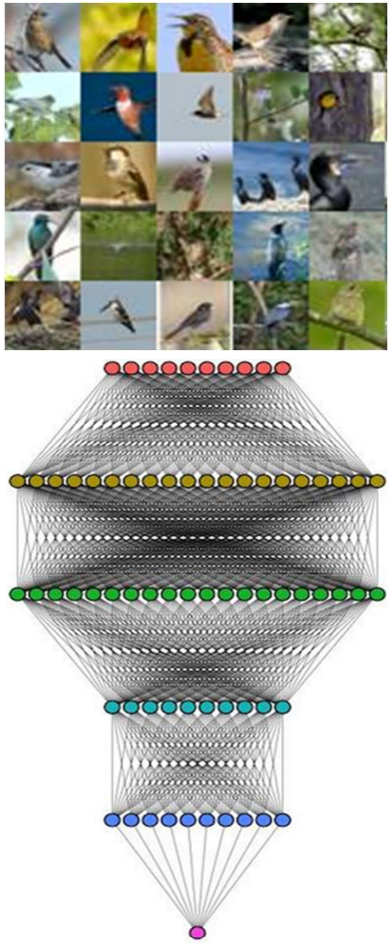
Discriminant Function

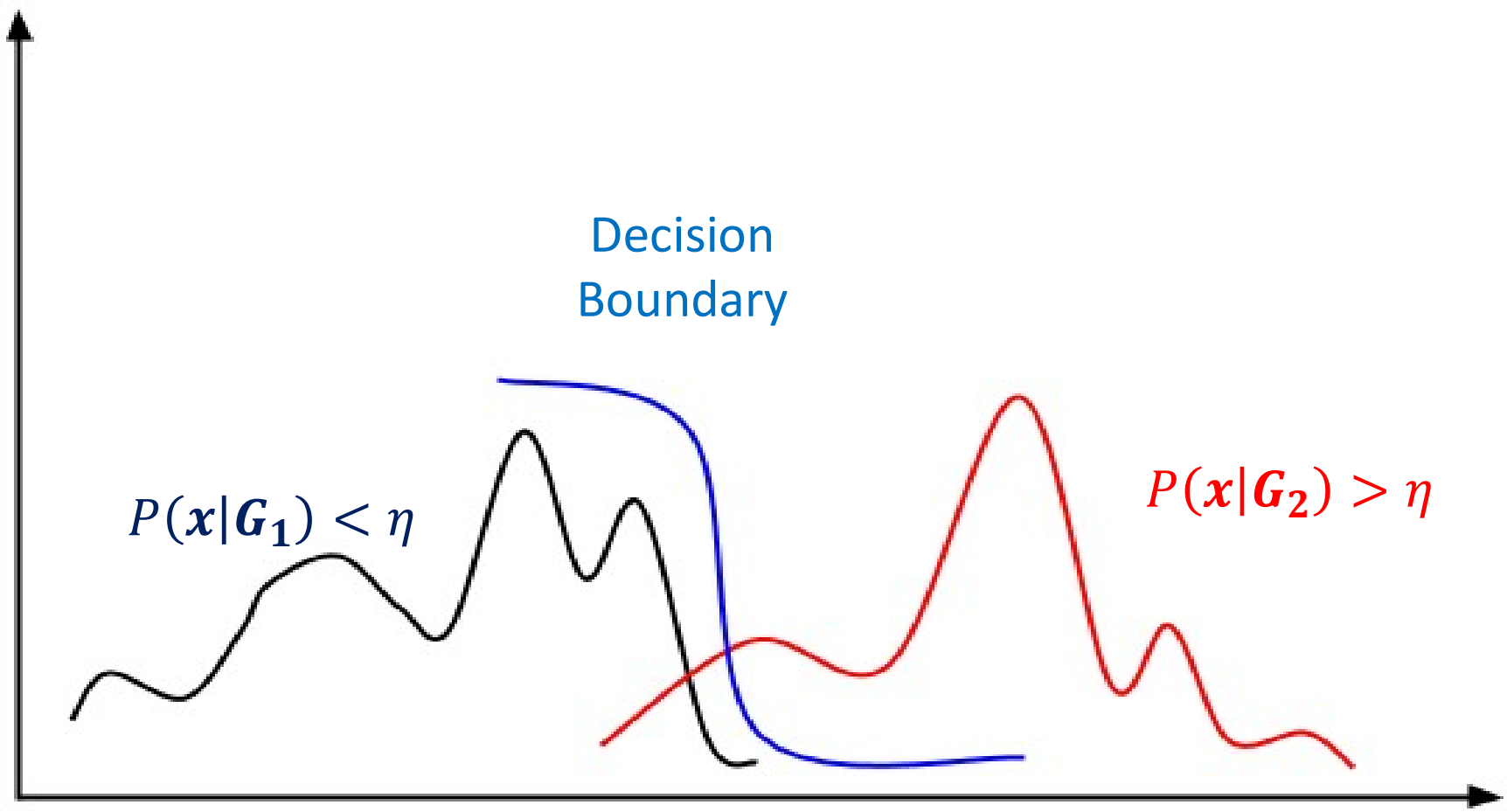
Prithwijit Guha
Dept. of EEE, IIT Guwahati

Supervised Learning

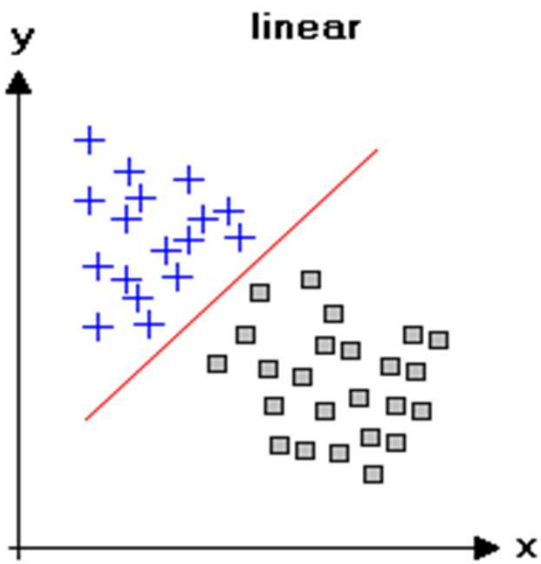


- Bayesian Classification, MAP, Chebyshev Inequality
- Performance Measures, Confusion Matrix, ROC Curves
- Discriminant Functions
- Logistic Regression
- Perceptron
- Multi-Layer Perceptron (MLP),
- Non-parametric Methods and K-NN
- Radial Basis Function Neural Networks
- Data Balancing; SMOTE & Weighted Loss Functions
- Classification & Regression Trees
- Support Vector Machines
- Ensemble Methods: Bagging and Boosting

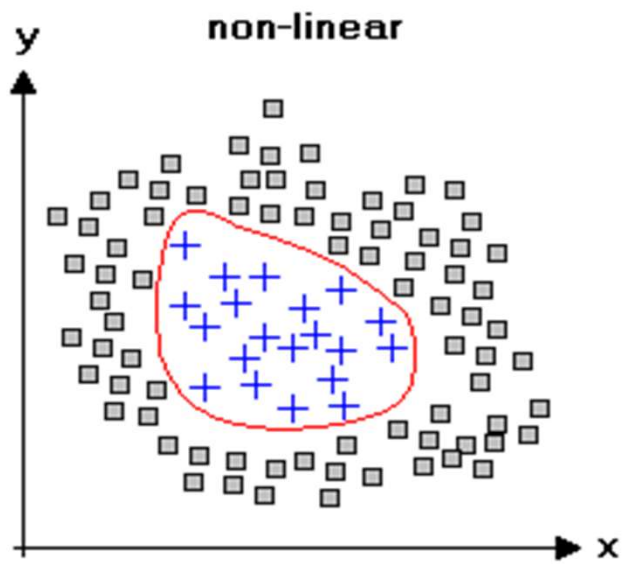
Discriminative Models vs. Generative Models



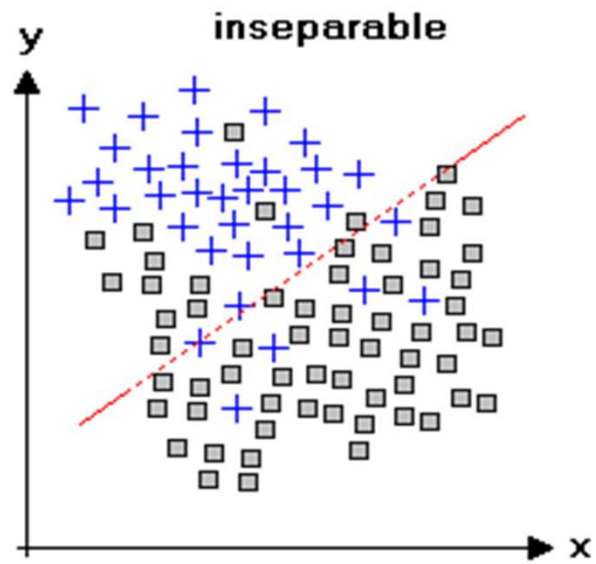
Separability in Feature Space



Linearly Separable



Separable by Non-Linear Boundary



Not Separable in Feature Space

Classification: Input Data & Label

$$\boldsymbol{D_0} = \{\boldsymbol{x}_i^0 : \boldsymbol{x}_i^0 \in \mathbb{R}^d ; i = 1, \dots n_0\}$$

$$\boldsymbol{D_1} = \{\boldsymbol{x}_j^1 : \boldsymbol{x}_j^1 \in \mathbb{R}^d ; j = 1, \dots n_1\}$$

$$y(\boldsymbol{x}) = \begin{cases} 1, & \boldsymbol{x} \in \boldsymbol{\omega_1} \\ 0, & \boldsymbol{x} \in \boldsymbol{\omega_0} \end{cases}$$

Classification: Input Distribution

$$\boldsymbol{x} \in \boldsymbol{\omega_0} \Rightarrow \boldsymbol{x} \sim P_0(\boldsymbol{x}; \boldsymbol{\theta}_0)$$

$$\boldsymbol{x} \in \boldsymbol{\omega_1} \Rightarrow \boldsymbol{x} \sim P_1(\boldsymbol{x}; \boldsymbol{\theta}_1)$$

P_0 and P_1 are the respective Probability Distributions learned from $\boldsymbol{D_0}$ and $\boldsymbol{D_1}$. The respective parameters of these Distributions are $\boldsymbol{\theta}_0$ and $\boldsymbol{\theta_1}$.

Classification: Input Distribution

$$P(\boldsymbol{\omega_1}|\boldsymbol{x}) = \frac{P(\boldsymbol{x}|\boldsymbol{\omega_1}, \boldsymbol{\theta_1})P(\boldsymbol{\omega_1})}{P(\boldsymbol{x})}$$

Posterior **Likelihood** **Prior**

$$P(\boldsymbol{\omega_0}|\boldsymbol{x}) = \frac{P(\boldsymbol{x}|\boldsymbol{\omega_0}, \boldsymbol{\theta_0})P(\boldsymbol{\omega_0})}{P(\boldsymbol{x})}$$

Evidence

$$P(\boldsymbol{x}) = P(\boldsymbol{x}|\boldsymbol{\omega_0}, \boldsymbol{\theta_0})P(\boldsymbol{\omega_0}) + P(\boldsymbol{x}|\boldsymbol{\omega_1}, \boldsymbol{\theta_1})P(\boldsymbol{\omega_1})$$

Discriminant Functions & Decision Rule

MAP Decision Rule

$$y(\boldsymbol{x}) = \begin{cases} 1, & P(\boldsymbol{\omega_1}|\boldsymbol{x}) > P(\boldsymbol{\omega_0}|\boldsymbol{x}) \\ 0, & P(\boldsymbol{\omega_1}|\boldsymbol{x}) < P(\boldsymbol{\omega_0}|\boldsymbol{x}) \end{cases}$$

Discriminant Function

$$g_i(\boldsymbol{x}) = \ln\{P(\boldsymbol{\omega_i}|\boldsymbol{x})\}$$

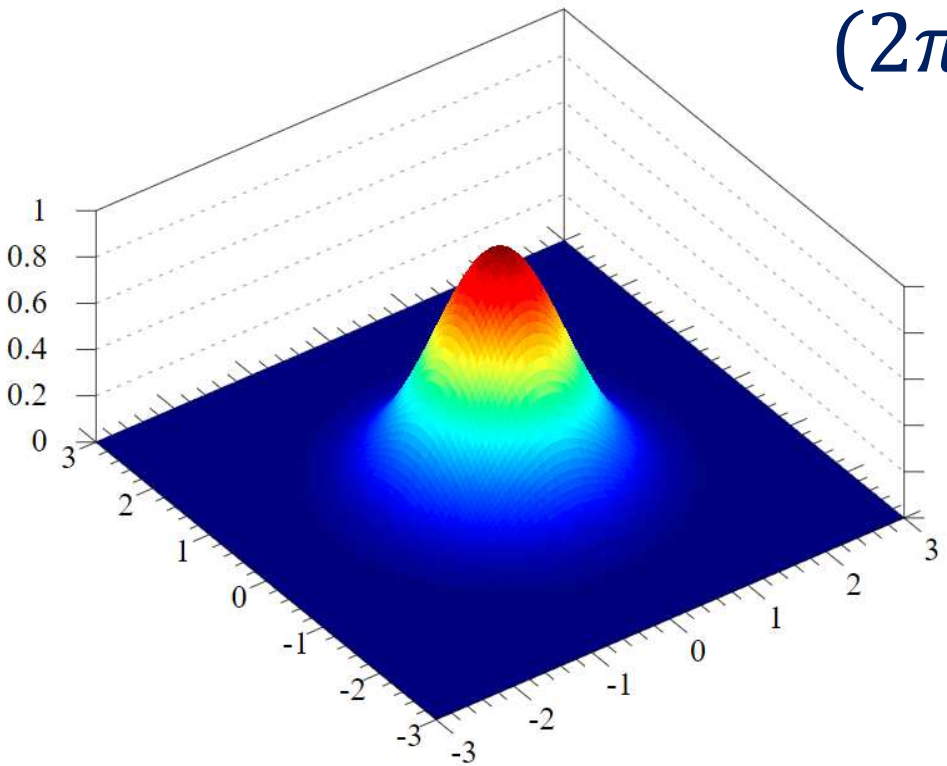
$$g(\boldsymbol{x}) = g_1(\boldsymbol{x}) - g_0(\boldsymbol{x})$$

$$\boxed{y(\boldsymbol{v}) = \begin{cases} 1, & g(\boldsymbol{v}) = g_1(\boldsymbol{v}) - g_0(\boldsymbol{v}) > 0 \\ 0, & g(\boldsymbol{v}) = g_1(\boldsymbol{v}) - g_0(\boldsymbol{v}) < 0 \end{cases}}$$

Classification Decision Rule (unseen data $\boldsymbol{v}$)

Discriminant Functions: Gaussian Distribution

$$P(\boldsymbol{x};\boldsymbol{\theta}=[\boldsymbol{\mu},\boldsymbol{C}])=\frac{1}{(2\pi)^{\frac{d}{2}}|\boldsymbol{C}|^{\frac{1}{2}}}e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}$$



Case-1: $\boldsymbol{C}_1=\boldsymbol{C}_2=\sigma^2\boldsymbol{I}$

Case-2: $\boldsymbol{C}_1=\boldsymbol{C}_2=\boldsymbol{C}$

Case-3: $\boldsymbol{C}_1\neq\boldsymbol{C}_2$

Discriminant Function: Gaussian Distribution

$$g_i(\boldsymbol{x}) = ln\{P(\boldsymbol{\omega_i}|\boldsymbol{x})\} = ?$$

Discriminant Function: Gaussian Distribution

$$g_i(\boldsymbol{x}) = ln\{P(\boldsymbol{\omega_i}|\boldsymbol{x})\} = ln\left\{\frac{P(\boldsymbol{x}|\boldsymbol{\omega_i}, \boldsymbol{\theta_i})P(\boldsymbol{\omega_i})}{P(\boldsymbol{x})}\right\}$$

$$= ln\{P(\boldsymbol{x}|\boldsymbol{\omega_i}, \boldsymbol{\theta_i})\} + ln\{P(\boldsymbol{\omega_i})\} - ln\{P(\boldsymbol{x})\}$$

Discriminant Function: Gaussian Distribution

$$ln\{P(\boldsymbol{x}|\boldsymbol{\omega_i},\boldsymbol{\theta_i})\} = \ln\left\{\frac{1}{(2\pi)^{\frac{d}{2}}|\boldsymbol{C_i}|^{\frac{1}{2}}}e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu_i})^T\boldsymbol{C_i}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_i})}\right\}$$

$$= -\frac{d}{2}ln\{2\pi\} - \frac{1}{2}ln\{|\boldsymbol{C}_i|\} - \frac{1}{2}(x-\boldsymbol{\mu}_i)^T\boldsymbol{C}_i^{-1}(x-\boldsymbol{\mu}_i)$$

Discriminant Function: Gaussian Distribution

$$g_i(\boldsymbol{x}) = -\frac{d}{2} ln\{2\pi\} - \frac{1}{2} ln\{|\boldsymbol{C}_i|\} - \frac{1}{2}(x - \boldsymbol{\mu}_i)^T \boldsymbol{C}_i^{-1}(x - \boldsymbol{\mu}_i)$$
$$+ ln\{P(\boldsymbol{\omega_i})\} - ln\{P(\boldsymbol{x})\}$$

Discriminant Function: Gaussian Distribution

$$g(\boldsymbol{x}) = g_1(\boldsymbol{x}) - g_0(\boldsymbol{x}) = ?$$

Discriminant Function: Gaussian Distribution

$$g_i(\boldsymbol{x}) = -\frac{d}{2} ln\{2\pi\} - \frac{1}{2} ln\{|\boldsymbol{C}_i|\} - \frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_i)^T \boldsymbol{C}_i^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_i) + ln\{P(\boldsymbol{\omega_i})\} - ln\{P(\boldsymbol{x})\}$$

$$\boxed{g(\boldsymbol{x}) = -\frac{1}{2} ln\left\{\frac{|\boldsymbol{C}_1|}{|\boldsymbol{C}_0|}\right\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}\{(\boldsymbol{x} - \boldsymbol{\mu}_1)^T \boldsymbol{C}_1^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_1) - (\boldsymbol{x} - \boldsymbol{\mu}_0)^T \boldsymbol{C}_0^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_0)\}}$$

Discriminant Function: $\boldsymbol{Case - I}$

$$C_0 = C_1 = \sigma^2 I$$

$$g(\boldsymbol{x}) = g_1(\boldsymbol{x}) - g_0(\boldsymbol{x}) = ?$$

Discriminant Function: $\boldsymbol{C_1} = \boldsymbol{C_0} = \sigma^2 \boldsymbol{I}$

$$g(\boldsymbol{x}) = -\frac{1}{2} ln\left\{\frac{|\boldsymbol{C}_1|}{|\boldsymbol{C}_0|}\right\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}\{(\boldsymbol{x}-\boldsymbol{\mu}_1)^T \boldsymbol{C}_1^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_1) - (\boldsymbol{x}-\boldsymbol{\mu}_0)^T \boldsymbol{C}_0^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_0)\}$$

$$= -\frac{1}{2} ln\left\{\frac{\sigma^2}{\sigma^2}\right\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}\{(\boldsymbol{x}-\boldsymbol{\mu}_1)^T (\sigma^{-2}\boldsymbol{I})\,(\boldsymbol{x}-\boldsymbol{\mu}_1) - (\boldsymbol{x}-\boldsymbol{\mu}_0)^T (\sigma^{-2}\boldsymbol{I})(\boldsymbol{x}-\boldsymbol{\mu}_0)\}$$

$$= 0 + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2\sigma^2}\{(\boldsymbol{x}-\boldsymbol{\mu}_1)^T\,(\boldsymbol{x}-\boldsymbol{\mu}_1) - (\boldsymbol{x}-\boldsymbol{\mu}_0)^T\,(\boldsymbol{x}-\boldsymbol{\mu}_0)\}$$

$$= -\frac{1}{2\sigma^2}\{(\boldsymbol{x}^T\boldsymbol{x} - 2\boldsymbol{\mu}_1^T\,\boldsymbol{x} + \boldsymbol{\mu}_1^T\boldsymbol{\mu}_1) - (\boldsymbol{x}^T\boldsymbol{x} - 2\boldsymbol{\mu}_0^T\,\boldsymbol{x} + \boldsymbol{\mu}_0^T\boldsymbol{\mu}_0)\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\}$$

$$= -\frac{1}{2\sigma^2}\{-2(\boldsymbol{\mu}_1-\boldsymbol{\mu}_0)^T\boldsymbol{x} + (\boldsymbol{\mu}_1^T\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T\boldsymbol{\mu}_0)\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\}$$

Discriminant Function: $\boldsymbol{C_1} = \boldsymbol{C_0} = \sigma^2 \boldsymbol{I}$

$$g(\boldsymbol{x}) = -\frac{1}{2\sigma^2}\{-2(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \boldsymbol{x} + (\boldsymbol{\mu}_1^T \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \boldsymbol{\mu}_0)\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\}$$

$$g(\boldsymbol{x}) = \frac{1}{\sigma^2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \boldsymbol{x} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2\sigma^2}(\boldsymbol{\mu}_1^T \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \boldsymbol{\mu}_0)$$

$$g(\boldsymbol{x}) = \boldsymbol{a}^T \boldsymbol{x} + b$$

$$\boldsymbol{a} = \frac{1}{\sigma^2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$$

$$b = ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2\sigma^2}(\boldsymbol{\mu}_1^T \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \boldsymbol{\mu}_0)$$

Comparison of Methods: Case-I

Method: Case-I	Model Size	MUL	ADD	NLIN	CMP
Bayesian Classifier					
Discriminant Function					

Solution by Bayes Classifier: Case-I

$$P(\boldsymbol{\omega}_i|\boldsymbol{x}) \propto P(\boldsymbol{x}|\boldsymbol{\omega}_i)P(\boldsymbol{\omega}_i)$$

$$P(\boldsymbol{x}|\boldsymbol{\omega}_i)P(\boldsymbol{\omega}_i) = \frac{P(\boldsymbol{\omega}_i)}{(2\pi)^{\frac{d}{2}}\sqrt{|C_i|}} \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_i)^T C_i^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_i)\right\}$$

$$= \beta_i \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_i)^T(\sigma^2 I)^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_i)\right\}$$

$$\beta_i = \frac{P(\boldsymbol{\omega}_i)}{\sigma(2\pi)^{\frac{d}{2}}}$$

$$= \beta_i \exp\{\gamma(\boldsymbol{x}-\boldsymbol{\mu}_i)^T(\boldsymbol{x}-\boldsymbol{\mu}_i)\}$$

$$\gamma = -\frac{1}{2\sigma^2}$$

Solution by Bayes Classifier: Case-I

$$P(\boldsymbol{\omega}_i|\boldsymbol{x}) \propto \beta_i \exp\{\gamma(\boldsymbol{x}-\boldsymbol{\mu}_i)^T(\boldsymbol{x}-\boldsymbol{\mu}_i)\} \qquad i = 1,2$$

Parameters: $\beta_1, \beta_2, \gamma, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2$ → Model Size: $2d + 3$

Operation	Multiplication	Addition	Non-Linearity	Comparison
$\boldsymbol{v}_i = \boldsymbol{x} - \boldsymbol{\mu}_i$	0	d	0	0
$u_i = \boldsymbol{v}_i^T \boldsymbol{v}_i$	d	d	0	0
$c_i = \gamma * u_i$	1	0	0	0
$l_i = \exp(c_i)$	0	0	1	0
$p_i = \beta_i * l_i$	1	0	0	0
Computations	$2 \times (d+2)$	$2 \times 2d$	2×1	1

TWO Evaluations for TWO Classes

Solution by Discriminant Function: Case-I

$$g(\boldsymbol{x}) = \boldsymbol{a}^T \boldsymbol{x} + b$$

Parameters: $\boldsymbol{a}, b$ → Model Size: $d + 1$

Operation	Multiplication	Addition	Non-Linearity	Comparison
$u = \boldsymbol{a}^T \boldsymbol{x}$	d	d	0	0
$g = u + b$	0	1	0	0
Computations	d	$d + 1$	0	1

Evaluated Only Once

Comparison of Methods: Case-I

Method: Case-I	Model Size	MUL	ADD	NLIN	CMP
Bayesian Classifier	$2d+3$	$2(d+2)$	$4d$	2	1
Discriminant Function	$d+1$	d	$d+1$	0	1

Case-I: Discriminant Function (Discriminative) over Bayesian Classifier (Generative)

- Half the model size
- Half the number of Multiplications
- One Fourth the number of Additions
- No Non-Linearity Computation!!!
- Both has One Comparison

Discriminant Function: $\boldsymbol{Case - II}$

$$C_0 = C_1 = C$$

$$g(x) = ?$$

Discriminant Function: $\boldsymbol{C_1 = C_0 = C}$

$$g(\boldsymbol{x}) = -\frac{1}{2} ln\left\{\frac{|\boldsymbol{C}_1|}{|\boldsymbol{C}_0|}\right\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}\{(\boldsymbol{x}-\boldsymbol{\mu}_1)^T \boldsymbol{C}_1^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_1) - (\boldsymbol{x}-\boldsymbol{\mu}_0)^T \boldsymbol{C}_0^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_0)\}$$

$$= -\frac{1}{2} ln\left\{\frac{|\boldsymbol{C}|}{|\boldsymbol{C}|}\right\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}\{(\boldsymbol{x}-\boldsymbol{\mu}_1)^T \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_1) - (\boldsymbol{x}-\boldsymbol{\mu}_0)^T \boldsymbol{C}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_0)\}$$

$$= -\frac{1}{2}\{(\boldsymbol{x}^T \boldsymbol{C}^{-1}\boldsymbol{x} - 2\boldsymbol{x}^T \boldsymbol{C}^{-1}\boldsymbol{\mu}_1 + \boldsymbol{\mu}_1^T \boldsymbol{C}^{-1}\boldsymbol{\mu}_1) - (\boldsymbol{x}^T \boldsymbol{C}^{-1}\boldsymbol{x} - 2\boldsymbol{x}^T \boldsymbol{C}^{-1}\boldsymbol{\mu}_0 + \boldsymbol{\mu}_0^T \boldsymbol{C}^{-1}\boldsymbol{\mu}_0)\}$$
$$+ ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\}$$

$$= -\frac{1}{2}\{-2\boldsymbol{x}^T \boldsymbol{C}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0) + (\boldsymbol{\mu}_1^T \boldsymbol{C}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \boldsymbol{C}^{-1}\boldsymbol{\mu}_0)\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\}$$

Discriminant Function: $\boldsymbol{C_1 = C_0 = C}$

$$g(\boldsymbol{x}) = -\frac{1}{2}\{-2\boldsymbol{x}^T\boldsymbol{C}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0) + (\boldsymbol{\mu}_1^T\boldsymbol{C}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T\boldsymbol{C}^{-1}\boldsymbol{\mu}_0)\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\}$$

$$g(\boldsymbol{x}) = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T\boldsymbol{C}^{-1}\boldsymbol{x} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}(\boldsymbol{\mu}_1^T\boldsymbol{C}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T\boldsymbol{C}^{-1}\boldsymbol{\mu}_0)$$

$$g(\boldsymbol{x}) = \boldsymbol{p}^T\boldsymbol{x} + q$$

$$\boldsymbol{p} = \boldsymbol{C}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$$

$$q = ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}(\boldsymbol{\mu}_1^T\boldsymbol{C}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T\boldsymbol{C}^{-1}\boldsymbol{\mu}_0)$$

Comparison of Methods: Case-II

Method: Case-II	Model Size	MUL	ADD	NLIN	CMP
Bayesian Classifier					
Discriminant Function					

Solution by Bayes Classifier: Case-II

$$P(\boldsymbol{\omega}_i|\boldsymbol{x}) \propto P(\boldsymbol{x}|\boldsymbol{\omega}_i)P(\boldsymbol{\omega}_i)$$

$$P(\boldsymbol{x}|\boldsymbol{\omega}_i)P(\boldsymbol{\omega}_i) = \frac{P(\boldsymbol{\omega}_i)}{(2\pi)^{\frac{d}{2}}\sqrt{|C|}} \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_i)^T C^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_i)\right\}$$

$$= \beta_i \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_i)^T C^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_i)\right\}$$

$$\beta_i = \frac{P(\boldsymbol{\omega}_i)}{(2\pi)^{\frac{d}{2}}\sqrt{|C|}}$$

$$= \beta_i \exp\{(\boldsymbol{x}-\boldsymbol{\mu}_i)^T D(\boldsymbol{x}-\boldsymbol{\mu}_i)\}$$

$$D = -\frac{1}{2}C^{-1}$$

Solution by Bayes Classifier: Case-II

$$P(\boldsymbol{\omega}_i|\boldsymbol{x}) \propto \beta_i \exp\{(\boldsymbol{x}-\boldsymbol{\mu}_i)^T D(\boldsymbol{x}-\boldsymbol{\mu}_i)\} \qquad i = 1,2$$

Parameters: $\beta_1, \beta_2, D, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2$ $\longrightarrow$ Model Size: $\mathrm{d}^2 + 2d + 2$

Operation	Multiplication	Addition	Non-Linearity	Comparison
$\boldsymbol{v_i} = \boldsymbol{x} - \boldsymbol{\mu}_i$	0	d	0	0
$\boldsymbol{u_i} = D * \boldsymbol{v_i}$	d^2	d^2	0	0
$c_i = \boldsymbol{v_i^T u_i}$	d	d	0	0
$l_i = \exp(c_i)$	0	0	1	0
$p_i = \beta_i * l_i$	1	0	0	0
Computations	$2(d^2+d+1)$	$2(d^2+2d)$	2	1

TWO Evaluations for TWO Classes

Solution by Discriminant Function: Case-II

$$g(\boldsymbol{x}) = \boldsymbol{p}^T \boldsymbol{x} + q$$

Parameters: $\boldsymbol{p}, q$ → Model Size: $d + 1$

Operation	Multiplication	Addition	Non-Linearity	Comparison
$u = \boldsymbol{p}^T \boldsymbol{x}$	d	d	0	0
$g = u + b$	0	1	0	0
Computations	d	$d + 1$	0	1

Single Evaluation Only

Comparison of Methods: Case-II

Method: Case-II	Model Size	MUL	ADD	NLIN	CMP
Bayesian Classifier	$d^2 + 2d + 2$	$2(d^2 + d + 1)$	$2(d^2 + 2d)$	2	1
Discriminant Function	$d + 1$	d	$d + 1$	0	1

Case-II: Discriminant Function (Discriminative) over Bayesian Classifier (Generative)

- Model Size: First over Second Order in terms of Dimensions
- Multiplications and Additions: First over Second Order in terms of Dimension
- No Non-Linearity Computation!!!
- Both has Single Comparison

Discriminant Function: $\boldsymbol{Case - III}$

$$C_0 \neq C_1$$

$$g(\boldsymbol{x}) = g_1(\boldsymbol{x}) - g_0(\boldsymbol{x}) = ?$$

Discriminant Function: $\boldsymbol{C_1 \neq C_0}$

$$g(\boldsymbol{x}) = -\frac{1}{2} ln\left\{\frac{|\boldsymbol{C}_1|}{|\boldsymbol{C}_0|}\right\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}\{(\boldsymbol{x}-\boldsymbol{\mu}_1)^T \boldsymbol{C}_1^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_1) - (\boldsymbol{x}-\boldsymbol{\mu}_0)^T \boldsymbol{C}_0^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_0)\}$$

$$= -\frac{1}{2}\{(\boldsymbol{x}^T \boldsymbol{C}_1^{-1}\boldsymbol{x} - 2\boldsymbol{x}^T \boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1 + \boldsymbol{\mu}_1^T \boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1) - (\boldsymbol{x}^T \boldsymbol{C}_0^{-1}\boldsymbol{x} - 2\boldsymbol{x}^T \boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0 + \boldsymbol{\mu}_0^T \boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0)\}$$
$$+ ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2} ln\left\{\frac{|\boldsymbol{C}_1|}{|\boldsymbol{C}_0|}\right\}$$

$$= -\frac{1}{2}\{\boldsymbol{x}^T(\boldsymbol{C}_1^{-1} - \boldsymbol{C}_0^{-1})\boldsymbol{x} - 2\boldsymbol{x}^T(\boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0) + (\boldsymbol{\mu}_1^T \boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0)\} + ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\}$$
$$- \frac{1}{2} ln\left\{\frac{|\boldsymbol{C}_1|}{|\boldsymbol{C}_0|}\right\}$$

Discriminant Function: $\boldsymbol{C_1 \neq C_0}$

$$g(\boldsymbol{x}) = -\frac{1}{2}\{\boldsymbol{x}^T(\boldsymbol{C}_1^{-1} - \boldsymbol{C}_0^{-1})\boldsymbol{x} - 2\boldsymbol{x}^T(\boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0) + (\boldsymbol{\mu}_1^T\boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T\boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0)\}$$
$$+ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}ln\left\{\frac{|\boldsymbol{C}_1|}{|\boldsymbol{C}_0|}\right\})$$

$$g(\boldsymbol{x}) = \left[ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}(\boldsymbol{\mu}_1^T\boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T\boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0) - \frac{1}{2}ln\left\{\frac{|\boldsymbol{C}_1|}{|\boldsymbol{C}_0|}\right\}\right] + \boldsymbol{x}^T(\boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0)$$
$$-\frac{1}{2}\boldsymbol{x}^T(\boldsymbol{C}_1^{-1} - \boldsymbol{C}_0^{-1})\boldsymbol{x}$$

Discriminant Function: $\boldsymbol{C_1 \neq C_0}$

$$g(\boldsymbol{x}) = \left[ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}(\boldsymbol{\mu}_1^T \boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0) - \frac{1}{2} ln\left\{\frac{|\boldsymbol{C}_1|}{|\boldsymbol{C}_0|}\right\}\right] + \boldsymbol{x}^T(\boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0)$$
$$-\frac{1}{2}\boldsymbol{x}^T(\boldsymbol{C}_1^{-1} - \boldsymbol{C}_0^{-1})\boldsymbol{x}$$

$$g(\boldsymbol{x}) = \boldsymbol{x}^{\boldsymbol{T}}\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}^T\boldsymbol{x} + c$$

$$\boldsymbol{A} = \boldsymbol{C}_1^{-1} - \boldsymbol{C}_0^{-1}$$

$$\boldsymbol{b} = \boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0$$

$$c = \left[ln\left\{\frac{P(\boldsymbol{\omega_1})}{P(\boldsymbol{\omega_0})}\right\} - \frac{1}{2}(\boldsymbol{\mu}_1^T \boldsymbol{C}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \boldsymbol{C}_0^{-1}\boldsymbol{\mu}_0) - \frac{1}{2} ln\left\{\frac{|\boldsymbol{C}_1|}{|\boldsymbol{C}_0|}\right\}\right]$$

Comparison of Methods: Case-III

Method: Case-III	Model Size	MUL	ADD	NLIN	CMP
Bayesian Classifier					
Discriminant Function					

Solution by Bayes Classifier: Case-III

$$P(\boldsymbol{\omega}_i|\boldsymbol{x}) \propto P(\boldsymbol{x}|\boldsymbol{\omega}_i)P(\boldsymbol{\omega}_i)$$

$$P(\boldsymbol{x}|\boldsymbol{\omega}_i)P(\boldsymbol{\omega}_i) = \frac{P(\boldsymbol{\omega}_i)}{(2\pi)^{\frac{d}{2}}\sqrt{|C_i|}} \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_i)^T C_i^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_i)\right\}$$

$$= \beta_i \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_i)^T C_i^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_i)\right\}$$

$$\beta_i = \frac{P(\boldsymbol{\omega}_i)}{(2\pi)^{\frac{d}{2}}\sqrt{|C_i|}}$$

$$= \beta_i \exp\{(\boldsymbol{x}-\boldsymbol{\mu}_i)^T D_i(\boldsymbol{x}-\boldsymbol{\mu}_i)\}$$

$$D_i = -\frac{1}{2}C_i^{-1}$$

Solution by Bayes Classifier: Case-III

$$P(\boldsymbol{\omega}_i|\boldsymbol{x}) \propto \beta_i \exp\{(\boldsymbol{x}-\boldsymbol{\mu}_i)^T D_i(\boldsymbol{x}-\boldsymbol{\mu}_i)\} \qquad i = 1,2$$

Parameters: $\beta_1, \beta_2, D_1, D_2, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2 \longrightarrow$ Model Size: $2\mathrm{d}^2 + 2d + 2$

Operation	Multiplication	Addition	Non-Linearity	Comparison
$\boldsymbol{v_i} = \boldsymbol{x} - \boldsymbol{\mu}_i$	0	d	0	0
$\boldsymbol{u_i} = D_i * \boldsymbol{v_i}$	d^2	d^2	0	0
$c_i = \boldsymbol{v_i^T u_i}$	d	d	0	0
$l_i = \exp(c_i)$	0	0	1	0
$p_i = \beta_i * l_i$	1	0	0	0
Computations	$2(d^2 + d + 1)$	$2(d^2 + 2d)$	2	1

TWO Evaluations for TWO Classes

Solution by Discriminant Function: Case-III

$$g(\boldsymbol{x}) = \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} + \boldsymbol{b}^T \boldsymbol{x} + c$$

Parameters: $\boldsymbol{A}$, $\boldsymbol{b}$, c → Model Size: $\mathrm{d}^2 + d + 1$

Operation	Multiplication	Addition	Non-Linearity	Comparison
$\boldsymbol{v} = \boldsymbol{A}\boldsymbol{x}$	d^2	d^2	0	0
$u_1 = \boldsymbol{x}^T \boldsymbol{v}$	d	d	0	0
$u_2 = \boldsymbol{b}^T \boldsymbol{x}$	d	d	0	0
$g = u_1 + u_2 + c$	0	3	0	0
Computations	$d^2 + 2d$	$d^2 + 2d + 3$	0	1

Single Evaluation Only

Comparison of Methods: Case-III

Method: Case-III	Model Size	MUL	ADD	NLIN	CMP
Bayesian Classifier	$2d^2 + 2d + 2$	$2(d^2 + d + 1)$	$2(d^2 + 2d)$	2	1
Discriminant Function	$d^2 + d + 1$	$d^2 + 2d$	$d^2 + 2d + 3$	0	1

Case-III: Discriminant Function (Discriminative) over Bayesian Classifier (Generative)

- Half the Model Size, although both Second Order in terms of Dimensions
- Multiplications/Additions: Almost Half, but both Second Order in terms of Dimensions
- No Non-Linearity Computation!!!
- Both has Single Comparison

Discriminant Functions

Consider the instances from two training datasets obtained from two categories. The instances in each dataset are normally distributed

$$S_0 = \left\{ \begin{bmatrix} -4 \\ -3 \end{bmatrix}, \begin{bmatrix} -5 \\ -3 \end{bmatrix}, \begin{bmatrix} -3 \\ -1 \end{bmatrix} \right\}$$

Class 0: ω_0

$$S_1 = \left\{ \begin{bmatrix} 5 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 6 \\ 7 \end{bmatrix} \right\}$$

Class 1: ω_1

(a) Construct the Discriminant Function $g(\boldsymbol{x})$

(b) Evaluate $g(\boldsymbol{x})$ at $\boldsymbol{x}_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and Classify $\boldsymbol{x}_0$

Discriminant Functions (Soln.)

$$S_0 = \left\{ \begin{bmatrix} -4 \\ -3 \end{bmatrix}, \begin{bmatrix} -5 \\ -3 \end{bmatrix}, \begin{bmatrix} -3 \\ -1 \end{bmatrix} \right\}$$

Class 0

$$\boldsymbol{\mu}_0 = \begin{bmatrix} -4.00 \\ -2.33 \end{bmatrix} \qquad P(\omega_0) = 0.5$$

$$C_0 = \begin{bmatrix} 0.6667 & 0.6667 \\ 0.6667 & 0.8889 \end{bmatrix}$$

$$S_1 = \left\{ \begin{bmatrix} 5 \\ 5 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \end{bmatrix}, \begin{bmatrix} 6 \\ 7 \end{bmatrix} \right\}$$

Class 1

$$\boldsymbol{\mu}_1 = \begin{bmatrix} 5.0000 \\ 5.6667 \end{bmatrix} \qquad P(\omega_1) = 0.5$$

$$C_1 = \begin{bmatrix} 0.6667 & 0.6667 \\ 0.6667 & 0.8889 \end{bmatrix}$$

$$P(\omega_0) = P(\omega_1) \qquad C_0 = C_1$$

Discriminant Functions (Soln.)

$$g(\boldsymbol{x}) = \boldsymbol{p}^T\boldsymbol{x} + q \qquad (C_0 = C_1)$$

$$\boldsymbol{p} = \boldsymbol{C}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)$$

$$q = ln\left\{\frac{P(\omega_1)}{P(\omega_0)}\right\} - \frac{1}{2}(\boldsymbol{\mu}_1^T\boldsymbol{C}^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T\boldsymbol{C}^{-1}\boldsymbol{\mu}_0)$$

$$q = \frac{1}{2}(\boldsymbol{\mu}_0^T\boldsymbol{C}^{-1}\boldsymbol{\mu}_0 - \boldsymbol{\mu}_1^T\boldsymbol{C}^{-1}\boldsymbol{\mu}_1) \quad [P(\omega_0) = P(\omega_1)\,]$$

Discriminant Functions (Soln.)

$$g(\boldsymbol{x}) = \boldsymbol{p}^T \boldsymbol{x} + q$$

$$q = -1.5$$

$$\boldsymbol{p} = \begin{bmatrix} 18.0 \\ -4.5 \end{bmatrix}$$

$$\boldsymbol{x}_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$g(\boldsymbol{x_0}) = 21$$

$$\boldsymbol{x}_0 \in \omega_1$$

Summary

- Generative and Discriminative Models
- Discriminative Function: Log of Ratio of Posteriors
- Binary Classification Problem
- Three Conditions on Covariance Matrices
- Condition Specific Discrimination Functions
- Bayes Classifier and Discriminant Functions
- Comparing Model Sizes and Computations

Thank You